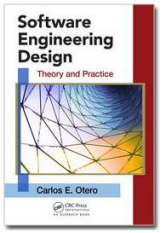
CHAPTER 10: MANAGEMENT, LEADERSHIP, AND ETHICS

-- Dr. L. Daniel Otero, Florida Institute of Technology --

SESSION I: DESIGN MANAGEMENT FRAMEWORK

Software Engineering Design: Theory and Practice
by Carlos E. Otero



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SESSION'S AGENDA

- Software Design Management
  - ✓ What is software design management?
  - ✓ Why study software design management?

- A Design Management Framework
  - ✓ Quality in software design
  - ✓ Typical project lifecycle
  - ✓ Typical performance measures in project phases
  - ✓ Components of the design management framework

- What's next…

SOFTWARE DESIGN MANAGEMENT

- What is software design management?
  - ✓ Software design is the process of transforming functional and non-functional requirements into models that describe a technical solution.
    - Can be viewed as a complex decision problem since
      - there are many ways to design a software system, and
      - software engineers must make design decisions to achieve the required goals of the software system in the most effective manner
  - ✓ Management is a *set of activities* directed at an organization's *resources* to achieve organizational goals in an efficient and effective manner.
    - Activities: planning and decision making, organizing, leading, and controlling
    - Resources: human, financial, physical, and information
  - ✓ In the design phase, management refers to activities required to efficiently create *quality* design artifacts, within schedule and budget constraints.

SOFTWARE DESIGN MANAGEMENT

- Why study software design management?
  - ✓ In large-scale software projects, software design management is essential to plan, organize, staff, track, and lead the activities required to carry out successfully the software architecture and detailed design steps.
  - ✓ To understand relevant managerial techniques and concepts
    - This will help to effectively use resources to achieve the various goals for each main component in the design phase (i.e., software architecture, detailed design, and documentation).
    - Effective use of resources is key to the success of software design management.
  - ✓ In the general sense, software design management is needed to:
    - Control/monitor design processes
    - Manage the resources necessary for completing quality design artifacts according to organizational goals

THE CONCEPT OF QUALITY

- Quality is a performance measure for a service provided or a product produced

- Quality is *relative* to a particular stakeholder. For example:
  - ✓ A personal website can be classified as high-quality by the programmer that developed it, and as low quality by user clients. Why?

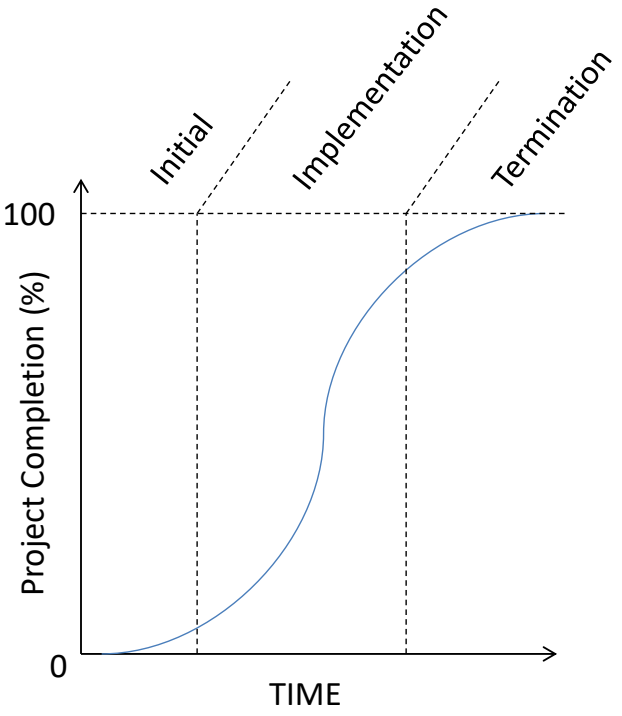


    - Because both stakeholders have different parameters by which they measure quality
      - The programmer may perceive quality in terms of being able to use appropriate colors and organize the website to run on a particular web browser.
      - A client using a different web browser may not see the website's organization and colors as intended; thus perceiving the website's quality as poor.

- Although the quality of a software design can be assessed in various ways, from a management's perspective, quality of software design tends to be evaluated in terms of *cost* and *scheduling*.
  - ✓ Thus, it is important to understand project management techniques that can be used to keep design artifacts within cost and schedule thresholds.

Design Management Framework

- It is beneficial to view the design phase as a project. Why?
  - ✓ Project management techniques can be directly applied to the design phase
    - This will help to efficiently and effectively manage the software design activities
    - E.g., the life cycle of the design phase can be represented with three stages as in:



- This life cycle represents a typical project pattern
  - ✓ Contributions to a project's completion are:
    - Relatively small during the initial and termination phases
    - Significantly high during the implementation phase
- When defining a life cycle structure for a project, we can decompose a project into its appropriate stages, and manage each stage individually.

LIFECYCLE STAGES

- In the initial stage, there are two main objectives:
  - ✓ Achieve a clear understanding of the tasks to be accomplished and the resources that are necessary to successfully complete the tasks.
  - ✓ Develop a plan to complete the project's tasks within schedule and budget constraints.

- The initial phase is characterized by planning activities.
  - ✓ Although some of the artifacts resulting from this phase (e.g., schedule) may be deliverables to clients, this phase is not viewed as contributing significantly to the percentage project completion.

- The implementation stage is where:
  - ✓ Most of the effort is employed
  - ✓ Most of the design project deliverables are achieved, including completed design documents
  - ✓ The percent project completion is the highest

Objectives in the Lifecycle Stages

- In the termination stage,
  - ✓ Key activities involve verification that everything is in place for a smooth transition into the code construction phase. Examples of activities are:
    - Making sure that the latest versions of the design documents are securely stored according to configuration management procedures
    - Updating schedule and cost current values
    - Re-evaluating schedule and budget plans based on the resulting performance measures of the design phase
    - Communicating results to upper management.

- The design documents generated in this stage are critical for the robustness and maintainability of the software designed.
  - ✓ However, the relative impact of this stage to project completion is not viewed as significant as that of the implementation phase.

MANAGING LIFECYCLE STAGES

- The lifecycle stages of a design project must be *individually managed* to ensure acceptable levels of quality and performance measures.
  - ✓ What does this means?
    - For example, an important outcome of the *planning stage* is a clear understanding of the tasks to be accomplished and the resources that are necessary to successfully complete the tasks.
      - – Various project management techniques exist to help achieve this outcome.

- It is important to know how to apply appropriate project management techniques to each of the design project stages
  - ✓ Afterwards, *peer reviews* will help ensure that the objectives of the planning stage are achieved.
    - Peer reviews must be conducted throughout these stages to minimize the propagation of errors to subsequent stages and apply corrective actions.



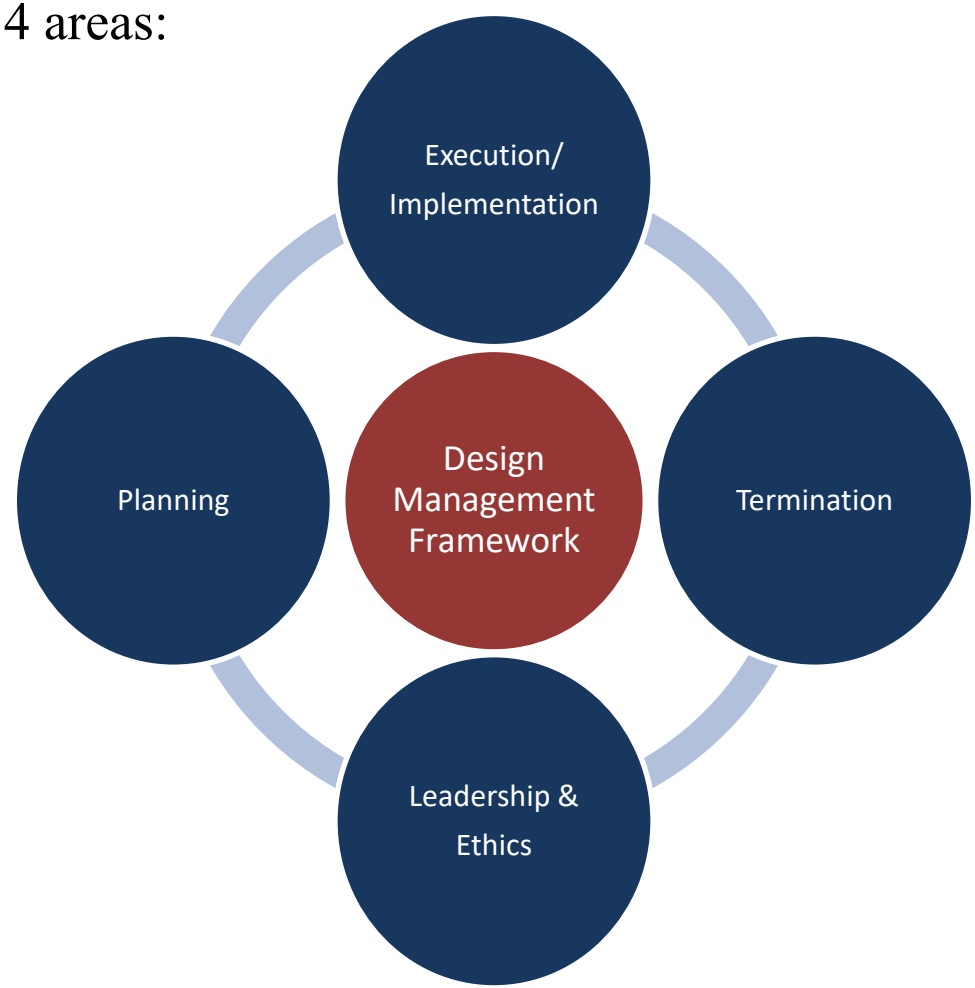
DESIGN MANAGEMENT FRAMEWORK

- Each of the design project lifecycle stages are associated with different performance measures. For example:
  - ✓ In the planning stage, performance measures such as the following are relevant:
    - Thoroughness (everything is considered),
    - Completeness (complete definition of plans),
    - Accuracy of estimates.
  - ✓ In the execution/implementation stage,
    - Quality is a function of a series of technical attributes of software designs.
    - Managing during this stage is mainly a monitoring approach to make sure that the plan is been followed, and take corrective actions if necessary.

- *Leadership* and *good ethical conduct* are important factors that affect the effectiveness of managing each lifecycle stage.

Design Management Framework

- Thus, an appropriate design management framework is one that considers the following 4 areas:



WHAT'S NEXT…

- This session accomplished the following:
  - ✓ Defined software design management
  - ✓ Understood the importance of studying software design management
  - ✓ Discussed the concept of quality in software design management
  - ✓ Described the stages of a typical project lifecycle
  - ✓ Described typical performance measures in project phases
  - ✓ Defined a design management framework composed of four components:
    - Planning
    - Implementation/Execution
    - Termination
    - Leadership & Ethics

- Next session will focus on the planning component of the design management framework,
  - ✓ This session will focus on managerial concepts and techniques that are relevant to the planning phase.

QUESTIONS?

